Lecture 14.

Q. Algorithms + Complexity Classes P, NP, QBP etc.

- First Q. Algorithm 1985 David Deutsch
- 1992 Deutsch-Jozsa

  (1st Q. Algo that's exponentially faster than any <u>deterministic</u> classical algorithm)

  but not faster than best probabilistic classical algo)

  (but small chance of failure)

- Simon Problem is in BQP (bounded-error Quantum Polynomial Time)

  ↳ Error bounded to 1/3.

  ↳ is exponentially faster than any classical algo, even including probabilistic ones.

These toy algorithms pave the way to Grover Search.

↳ Reduces complexity from

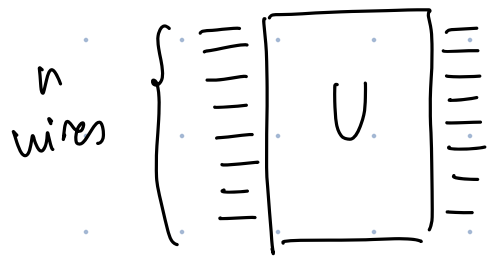
$$O(N) \rightarrow O(\sqrt{N})$$

Shor's Algorithm.

↳ good at solving hard problem that are easy to check (quadratic speed up)

Gadgets

① Phase 'Kickback' of a controlled unitary



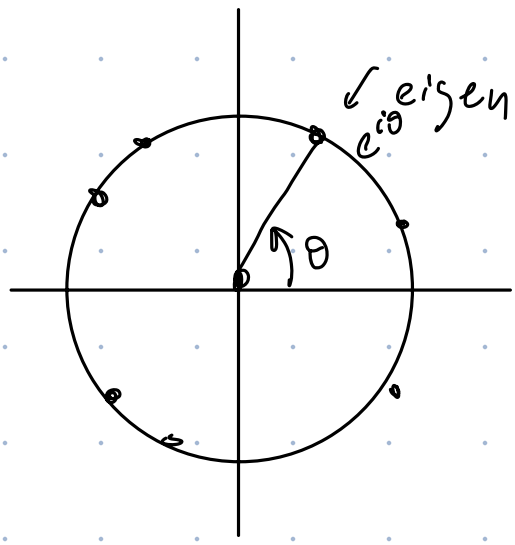
2^N to 2^N

Spectral Representation of $U = \sum_{j=0}^{2^N-1} e^{i\theta_j} |\psi_j\rangle\langle\psi_j|$, $\theta_j \in \mathbb{R}$.

j is an integer, and $|j\rangle$ represents a bit string $\underbrace{|b_n b_{n-1} \cdots b_2 b_1 b_0\rangle}_{\text{representation of } j}$.

$$U^\dagger = \sum_{j=0}^{2^N-1} e^{-i\theta_j} |\psi_j\rangle\langle\psi_j|$$

$UU^\dagger = I \Rightarrow$ unitary

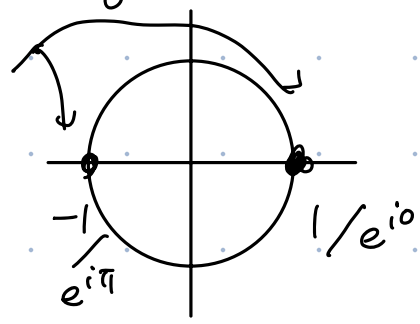


example: $U = Z$

$|\hat{n}\rangle\langle\hat{n}| - |-\hat{n}\rangle\langle-\hat{n}|$

$= \hat{n}\cdot\vec{\sigma}$

2 eigenvalues.

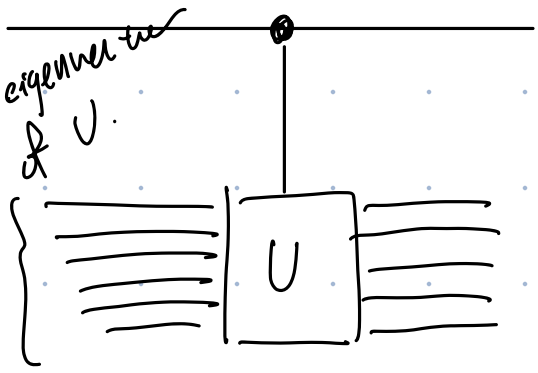


Controlled unitaries cU : cX = control NOT

$\alpha|0\rangle + \beta|1\rangle$

special case $|\psi_j\rangle$ is an eigenvector of U.

$|\psi_j\rangle$



apply I if $q_0 = 0$ &
apply U if $q_0 = 1$.

$$h = \sum_j \epsilon_j |\psi_j\rangle\langle\psi_j|$$

$\forall U \; \exists$ an h s.t. $U = e^{ih}$

$\hookrightarrow h = \sum_j \theta_j |\psi_j\rangle\langle\psi_j|$

$$|\Psi_j\rangle = |\psi_j\rangle \otimes (\alpha|0\rangle + \beta|1\rangle)$$

$$= |\psi_j\rangle \otimes \alpha|0\rangle + |\psi_j\rangle \otimes \beta|1\rangle$$

$$cU|\Psi\rangle = I|\psi_j\rangle \otimes \alpha|0\rangle + \underbrace{U|\psi_j\rangle}_{e^{i\theta_j}|\psi_j\rangle} \otimes \beta|1\rangle$$

$\rightarrow$ special case $|\psi_j\rangle$ is an eigenvector of U.

$|\psi\rangle = |\psi_1\rangle + |\psi_2\rangle$

$\downarrow$

$\underline{e^{i\theta_1}|\psi_1\rangle} + \underline{e^{i\theta_2}}|\psi_2\rangle$

global phase of individual eigenstates becomes relative phase for

$$= \underbrace{|\psi_j\rangle}_{\text{target didn't change}} \otimes \underbrace{(\alpha|0\rangle + \beta e^{i\theta_j}|1\rangle)}_{\text{control changes}}$$

$\hookrightarrow$ phase kickback onto the control.

how can you actually control the target without getting a phase kickback on the control.

same state.

$X|+\rangle = +|+\rangle$

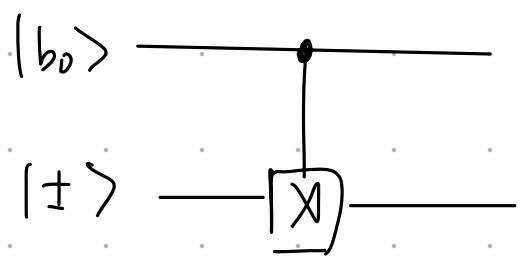
global phase.

$X|-\rangle = -|-\rangle$

same state

flipping a state that's sort of $|0\rangle$ & sort of $|1\rangle$ gives you back a state that is also sort of $|0\rangle$ and sort of $|1\rangle$ $\Rightarrow$ the same state.

$CNOT_0$



$b_0 = 0, 1$.

$$CNOT_0 |\pm\rangle \otimes |b_0\rangle = (\pm 1)^{b_0} |\pm\rangle \otimes |b_0\rangle$$

the sign of the global phase depends on the state of the control.

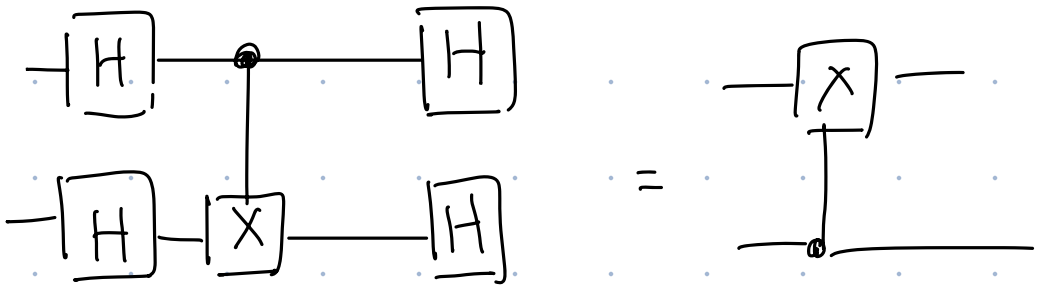
$$CNOT_0 |+\rangle \otimes |+\rangle = |+\rangle \otimes |+\rangle$$

$$CNOT_0 |-\rangle \otimes |+\rangle = |-\rangle \otimes |-\rangle$$

$$|-\rangle \otimes \left(\frac{|0\rangle + |1\rangle}{\sqrt{2}}\right)$$

$$\frac{|-\rangle \otimes |0\rangle}{\sqrt{2}} + \frac{-|-\rangle \otimes |1\rangle}{\sqrt{2}}$$

$$|-\rangle \otimes \left(\frac{|0\rangle - |1\rangle}{\sqrt{2}} \right)$$

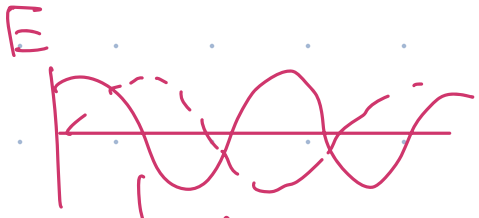


$(H \otimes H)\, CNOT_0\, (H \otimes H) = CNOT_1$

whose the control & whose the target depends on the basis!

→ basis dependent.

→ Use this trick in certain algorithm.



→ if you change the phase of

$\Delta\phi = 90^\circ \Rightarrow$ Y rotation.